

Texas Geometry Credit by Exam

Practice Exam B · 22 Questions

McGraw Hill Glencoe Geometry (Texas) 2015 · TEKS Aligned

Name: _____	Date: _____
Score: _____ / 22	Time Allowed: _____ min

DIRECTIONS: Select the best answer for each question. Each question is worth 1 point. Show all work in the space below each question. Calculators may be used unless otherwise noted. This exam is aligned to the Texas Essential Knowledge and Skills (TEKS) for Geometry.

[Triangle Similarity]

1. Two triangles are similar. The sides of the smaller triangle are 3, 4, and 5. The longest side of the larger triangle is 15. What is the scale factor from the smaller to the larger triangle?

- A. 2
- B. 3
- C. 4
- D. 5

Work space: _____

[Dilations]

2. A point at (4, -6) is dilated by a scale factor of $\frac{1}{2}$ centered at the origin. What are the coordinates of the image?

- A. (8, -12)
- B. (2, -3)
- C. (4, -3)
- D. (2, -6)

Work space: _____

[Trigonometric Ratios]

3. In a right triangle, angle theta is opposite a leg of length 8 and adjacent to a leg of length 15. What is $\cos(\theta)$?

- A. $\frac{8}{17}$
- B. $\frac{15}{17}$
- C. $\frac{8}{15}$
- D. $\frac{15}{8}$

Work space: _____

[Parallelogram Properties]

4. In parallelogram ABCD, the diagonals intersect at point E. If $AC = 28$, what is AE ?

- A. 7
- B. 14
- C. 28
- D. 56

Work space: _____

[Rectangle Properties]

5. A rectangle has a diagonal of length 13 and one side of length 5. What is the length of the other side?

- A. 8
- B. 10
- C. 12
- D. 15

Work space: _____

[Circles – Arc Length]

6. A circle has radius 6. What is the arc length corresponding to a 90° central angle? (Give an exact answer.)

- A. 3π
- B. 6π
- C. 9π
- D. 12π

Work space: _____

[Circles – Inscribed Angle Theorem]

7. An inscribed angle in a circle intercepts an arc of 140° . What is the measure of the inscribed angle?

- A. 35°
- B. 70°
- C. 140°
- D. 280°

Work space: _____

[Circles – Secant-Tangent Relationship]

8. A tangent segment from an external point to a circle has length 12. A secant from the same point has an external segment of length 4. What is the total length of the secant?

- A. 9
- B. 16
- C. 36
- D. 48

Work space: _____

[Circles – Area]

9. What is the area of a circle with diameter 10?

- A. 10π
- B. 20π
- C. 25π
- D. 100π

Work space: _____

[Surface Area – Rectangular Prism]

10. What is the total surface area of a rectangular prism with length 5, width 3, and height 4?

- A. 47
- B. 60
- C. 94
- D. 120

Work space: _____

[Volume – Cylinder]

11. What is the volume of a cylinder with radius 3 and height 10?

- A. 30π
- B. 60π
- C. 90π
- D. 120π

Work space: _____

[Transformations – Reflections]

12. A point at (3, -5) is reflected over the x-axis. What are the coordinates of the image?

- A. (3, 5)
- B. (-3, 5)
- C. (-3, -5)
- D. (5, -3)

Work space: _____

[Transformations – Translations]

13. A point at (2, 4) is translated by the vector (-5, 3). What are the coordinates of the image?

- A. (-3, 7)
- B. (7, 1)
- C. (-3, 1)
- D. (7, 7)

Work space: _____

[Trapezoid – Area]

14. A trapezoid has parallel sides of length 8 and 14, and a height of 6. What is the area?

- A. 48
- B. 66
- C. 88
- D. 132

Work space: _____

[Rhombus Properties]

15. A rhombus has diagonals of lengths 10 and 24. What is its perimeter?

- A. 26
- B. 52
- C. 60
- D. 104

Work space: _____

[Similar Triangles – Proportional Parts]

16. In triangle ABC, segment DE is parallel to BC, with D on AB and E on AC. AD = 4, DB = 8, and DE = 5. What is BC?

- A. 10
- B. 12
- C. 15
- D. 20

Work space: _____

[Volume – Pyramid]

17. What is the volume of a square pyramid with base side length 6 and height 9?

- A. 54
- B. 108
- C. 162
- D. 324

Work space: _____

[Coordinate Geometry – Triangle Area]

18. A triangle has vertices at (0, 0), (6, 0), and (0, 4). What is the area of the triangle?

- A. 10
- B. 12
- C. 16
- D. 24

Work space: _____

[Transformations – Rotations]

19. A point at (3, 4) is rotated 90° counterclockwise about the origin. What are the coordinates of the image?

- A. (4, -3)
- B. (-4, 3)
- C. (-3, 4)
- D. (3, -4)

Work space: _____

[Circles – Intersecting Chords Theorem]

20. Two chords intersect inside a circle. One chord is divided into segments of length 4 and 9. The other chord is divided into segments of 6 and x . What is x ?

- A. 4
- B. 6
- C. 9
- D. 36

Work space: _____

[Regular Polygons – Interior Angles]

21. What is the measure of each interior angle of a regular octagon?

- A. 45°
- B. 120°
- C. 135°
- D. 150°

Work space: _____

[Geometric Probability]

22. A point is chosen at random inside a square with side length 8. A circle of radius 2 is centered at the center of the square. What is the probability the point lands inside the circle?

- A. $\pi/16$
- B. $\pi/8$
- C. $\pi/4$
- D. $\pi/2$

Work space: _____

Answer Key – Practice Exam B

#	Answer	#	Answer
1	B	12	A
2	B	13	A
3	B	14	B
4	B	15	B
5	C	16	C
6	A	17	B
7	B	18	B
8	C	19	B
9	C	20	B
10	C	21	C
11	C	22	A

TEKS Standards Covered

Q #	Topic	TEKS
1	Triangle Similarity – Scale Factor	G.7(A)
2	Dilations – Coordinate Rule	G.3(C)
3	Trigonometric Ratios (SOH-CAH-TOA)	G.9(A)
4	Parallelogram – Diagonals Bisect	G.6(E)
5	Rectangle – Diagonal Length	G.6(E)
6	Circle – Arc Length	G.12(B)
7	Circle – Inscribed Angle Theorem	G.12(A)
8	Circle – Secant-Tangent Theorem	G.12(A)
9	Circle – Area	G.11(B)
10	Surface Area – Rectangular Prism	G.11(C)
11	Volume – Cylinder	G.11(D)
12	Transformations – Reflections	G.3(A)
13	Transformations – Translations	G.3(A)
14	Trapezoid – Area Formula	G.11(B)
15	Rhombus – Diagonals and Perimeter	G.6(E)
16	Similar Triangles – Proportional Parts	G.7(B)
17	Volume – Pyramid	G.11(D)
18	Coordinate Geometry – Triangle Area	G.2(B)

19	Transformations – Rotations	G.3(A)
20	Circle – Intersecting Chords Theorem	G.12(A)
21	Regular Polygons – Interior Angle Measure	G.6(A)
22	Geometric Probability	G.13(B)

Detailed Explanations – Exam B

1. (B) [Triangle Similarity]

Scale factor = (corresponding side of larger) / (corresponding side of smaller) = $15 / 5 = 3$.

2. (B) [Dilations]

For a dilation centered at the origin with scale factor k : $(x, y) \rightarrow (kx, ky)$.

$(4 \cdot 1/2, -6 \cdot 1/2) = (2, -3)$.

3. (B) [Trigonometric Ratios]

Find hypotenuse: $c = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$.

$\cos(\theta) = \text{adjacent} / \text{hypotenuse} = 15/17$.

4. (B) [Parallelogram Properties]

The diagonals of a parallelogram **bisect each other**, so E is the midpoint of AC. $AE = AC / 2 = 28 / 2 = 14$.

5. (C) [Rectangle Properties]

In a rectangle, the diagonal is the hypotenuse of a right triangle formed by the two sides.

$d^2 = l^2 + w^2 \rightarrow 169 = 25 + w^2 \rightarrow w^2 = 144 \rightarrow w = 12$.

6. (A) [Circles – Arc Length]

Arc length = (central angle / 360°) $\times 2\pi \cdot r = (90/360) \times 2\pi(6) = (1/4) \times 12\pi = 3\pi$.

7. (B) [Circles – Inscribed Angle Theorem]

By the **Inscribed Angle Theorem**, an inscribed angle is half the measure of its intercepted arc.

Inscribed angle = $140^\circ / 2 = 70^\circ$.

8. (C) [Circles – Secant-Tangent Relationship]

Tangent-Secant Theorem: $(\text{tangent})^2 = (\text{external part of secant}) \times (\text{whole secant})$.

$12^2 = 4 \times \text{whole secant} \rightarrow 144 = 4 \times \text{whole secant} \rightarrow \text{whole secant} = 36$.

9. (C) [Circles – Area]

Diameter = 10 $\rightarrow$ radius $r = 5$.

Area = $\pi \cdot r^2 = \pi \cdot 25 = 25\pi$.

10. (C) [Surface Area – Rectangular Prism]

$SA = 2(lw + lh + wh)$

$= 2[(5)(3) + (5)(4) + (3)(4)]$

$= 2[15 + 20 + 12] = 2(47) = 94$.

11. (C) [Volume – Cylinder]

$V = \pi \cdot r^2 \cdot h = \pi \cdot (3^2) \cdot 10 = \pi \cdot 9 \cdot 10 = 90\pi$.

12. (A) [Transformations – Reflections]

Reflecting over the **x-axis**: $(x, y) \rightarrow (x, -y)$.

$(3, -5) \rightarrow (3, 5)$.

13. (A) [Transformations – Translations]

Add the vector to the coordinates:

$$(2 + (-5), 4 + 3) = (-3, 7).$$

14. (B) [Trapezoid – Area]

$$\begin{aligned}\text{Area of trapezoid} &= (1/2) \cdot h \cdot (b_1 + b_2) \\ &= (1/2)(6)(8 + 14) = 3 \cdot 22 = 66.\end{aligned}$$

15. (B) [Rhombus Properties]

The diagonals of a rhombus bisect each other at right angles, forming right triangles with legs 5 and 12.

$$\text{Side} = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$$

$$\text{Perimeter} = 4 \times 13 = 52.$$

16. (C) [Similar Triangles – Proportional Parts]

By AA Similarity, triangle ADE is similar to triangle ABC.

$$\text{Ratio} = AD / AB = 4 / (4 + 8) = 4/12 = 1/3.$$

$$DE / BC = 1/3 \rightarrow BC = 3 \times 5 = 15.$$

17. (B) [Volume – Pyramid]

$$V = (1/3) \times B \times h, \text{ where } B = \text{base area} = 6^2 = 36.$$

$$V = (1/3) \times 36 \times 9 = (1/3) \times 324 = 108.$$

18. (B) [Coordinate Geometry – Triangle Area]

The base lies along the x-axis with length 6; the height is along the y-axis with length 4.

$$\text{Area} = (1/2) \times \text{base} \times \text{height} = (1/2)(6)(4) = 12.$$

19. (B) [Transformations – Rotations]

For a 90° counterclockwise rotation about the origin: $(x, y) \rightarrow (-y, x)$.

$$(3, 4) \rightarrow (-4, 3) = (-4, 3).$$

20. (B) [Circles – Intersecting Chords Theorem]

Intersecting Chords Theorem: (segment 1)(segment 2) = (segment 3)(segment 4).

$$4 \times 9 = 6 \times x \rightarrow 36 = 6x \rightarrow x = 6.$$

21. (C) [Regular Polygons – Interior Angles]

Each interior angle of a regular n-gon = $(n - 2) \times 180^\circ / n$.

$$\text{For } n = 8: (8 - 2) \times 180^\circ / 8 = 6 \times 180^\circ / 8 = 1080^\circ / 8 = 135^\circ.$$

22. (A) [Geometric Probability]

$$P = (\text{area of circle}) / (\text{area of square})$$

$$= \pi(2^2) / (8^2)$$

$$= 4\pi / 64 = \pi/16.$$